Solution

In the above pattern of polygons, in the first shape ($x = 1$), we get a closed shape called a triangle. Similarly the second shape ($x = 2$) gives a four sided polygon and the third shape ($x = 3$) is a five sided polygon and continuing in the same way two more shapes are formed. If the number of match sticks required to form each of the shapes is taken as y, then the values of x and y are tabulated as given below.

x	1	2	3	4	5	...
y	3	4	5	6	7	...

Observe the table and express y in terms of x as below :

when $x = 1$, $y = 3 = 1+2$

when $x = 2$, $y = 4 = 2+2$

when $x = 3$, $y = 5 = 3+2$

when $x = 4$, $y = 6 = 4+2$

when $x = 5$, $y = 7 = 5+2$

Hence, each of the values of y which we get from the table is 2 more than x. That is $y = x + 2$.

Therefore,

6th shape (x=6) will have $y = 8 = 6+2$ (8 match sticks)

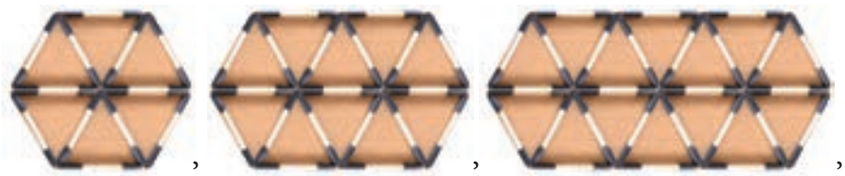
7th shape (x=7) will have $y = 9 = 7+2$ (9 match sticks)

8th shape (x=8) will have $y = 10 = 8+2$ (10 match sticks)

We clearly see that the next three shapes will require 8, 9 and 10 matchsticks.

Activity

Observe the pattern given below. Continue the pattern for three more steps.



, , , ___ , ___ , ___ .

Let, 'x' be the number of steps and 'y' be the number of match sticks. Tabulate the values of 'x' and 'y' and verify the relationship $y = 7x+5$.

7th_Maths_T2_EM_Chp5_1.indd 91 09-07-2019 12:59:53